

Assignment 10

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CSCI 4100

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1 Excercise 6.1

1.1 (a)

High Cosine Similarity and Low Euclidean Similarity

Consider two vectors:

- Vector 1 = $[1, 2, 3]$
- Vector 2 = $[1000, 2000, 3000]$

These two vectors are aligned in the same direction, resulting in a cosine similarity of 1, indicating they point in exactly the same direction (0 degrees apart). However, due to the difference in their magnitudes (Vector 2 is much longer than Vector 1), their Euclidean distance is relatively large (3738). This large Euclidean distance indicates a low Euclidean similarity between them, as Euclidean similarity decreases with distance.

Low Cosine Similarity and High Euclidean Similarity

Now consider:

- Vector 3 = $[0.1, 0.2, 0.3]$
- Vector 4 = $[-0.0001, -0.0002, -0.0003]$

These two vectors are oriented in exactly opposite directions, so their cosine similarity is -1 , showing they are pointing in opposite directions (180 degrees apart). However, since both vectors are quite short, the Euclidean distance between them is small (0.3745), indicating a relatively high Euclidean similarity.

Note: In both examples, the vectors originate from the same point, allowing us to focus on direction and magnitude.

1.2 (b)

Cosine similarity is primarily affected by the angle between vectors rather than their magnitudes or origin. Therefore, if the origin shifts, the cosine

similarity remains mostly unaffected, as it measures orientation rather than absolute position. However, Euclidean distance, which measures the direct distance between points, is sensitive to changes in origin and would be impacted by a shift.

2 Excercise 6.2

Solution to Exercise 6.2

Let

$$f(x) = \begin{cases} +1, & \text{if } \pi(x) \geq \frac{1}{2} \\ -1, & \text{otherwise.} \end{cases}$$

We want to show that

$$e(f(x)) = \mathbb{P}(f(x) \neq y) = \min\{\pi(x), 1 - \pi(x)\} \quad \text{and} \quad e(f(x)) \leq e(h(x)) \text{ for any other } h.$$

Starting with the error probability $e(f(x))$:

$$\begin{aligned} e(f(x)) &= \mathbb{P}(f(x) \neq y) \\ &= \mathbb{P}(\pi(x) \neq y) \\ &= \mathbb{P}(f(x) = +1, y = -1) + \mathbb{P}(f(x) = -1, y = +1) \\ &= \frac{1}{2}\pi(x) \times (1 - \pi(x)) + (1 - \pi(x)) \times \frac{1}{2}\pi(x) \\ &= \pi(x) \times (1 - \pi(x)) \end{aligned}$$

Thus,

$$e(f(x)) = \min\{\pi(x), (1 - \pi(x))\}$$

Case Analysis: When $\pi(x) \geq \frac{1}{2}$, we have $f(x) = 1$, and the error $e(f(x)) = 1 - \pi(x)$, while $\pi(x) \geq 1 - \pi(x)$, which means:

$$e(f(x)) = \min(\pi(x), 1 - \pi(x)).$$

Similarly, when $\pi(x) < \frac{1}{2}$, we get $f(x) = -1$, and $e(f(x)) = \pi(x)$, with $\pi(x) < 1 - \pi(x)$, leading to:

$$e(f(x)) = \min(\pi(x), 1 - \pi(x)).$$

This proves that $e(f(x)) = \min(\pi(x), 1 - \pi(x))$ holds for all values of $\pi(x)$.

Since $e(f(x))$ represents the minimal error that captures only irreducible stochastic noise, any other hypothesis differing from $f(x)$ would only increase errors rather than decrease them.

3 Problem 6.1

3.1 a

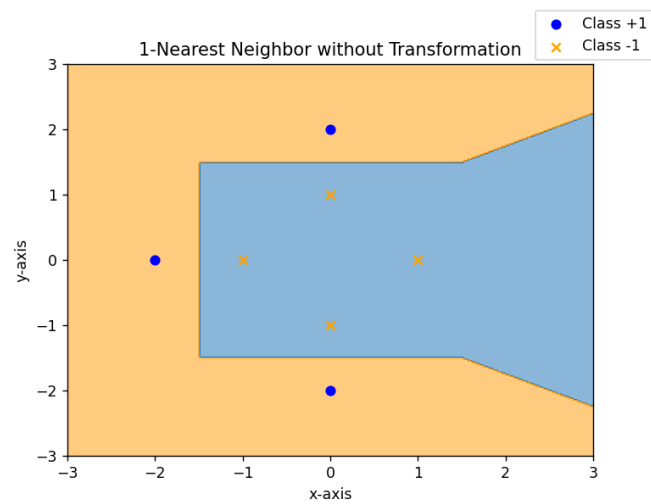


Figure 1: 1-NN without transform

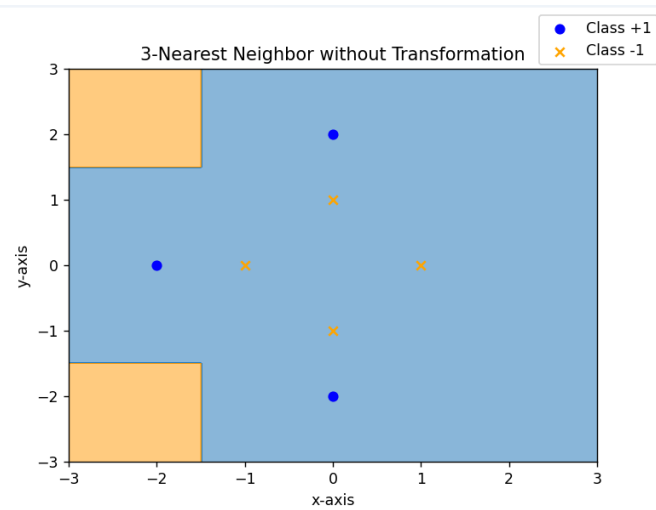


Figure 2: 3-NN without transform

3.2 b

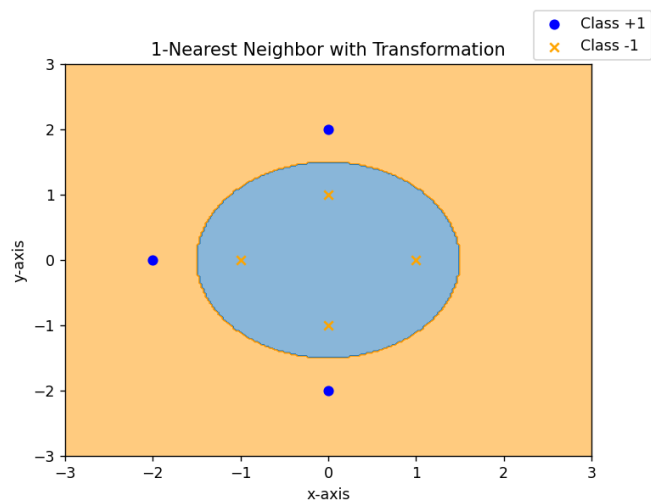


Figure 3: 1-NN with transform

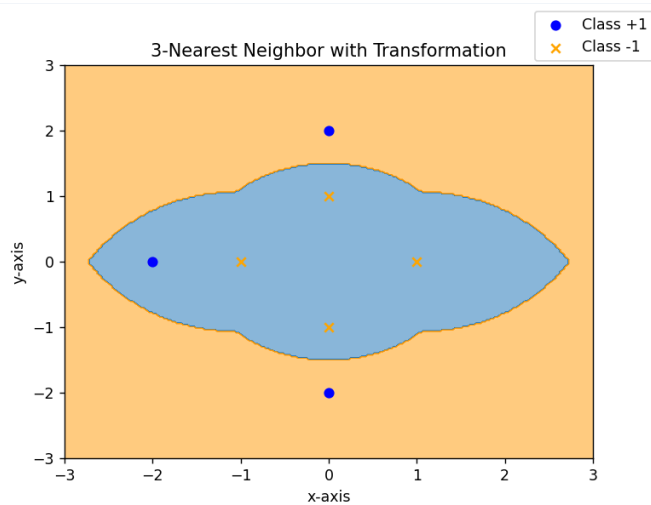


Figure 4: 3-NN with transform

4 Problem 6.4

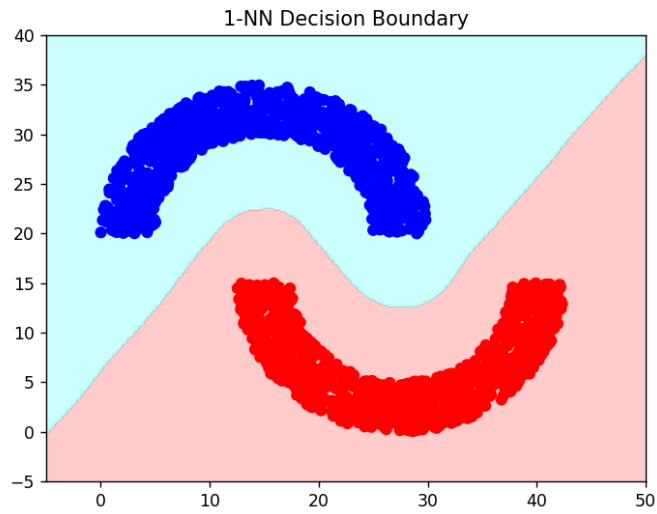


Figure 5: 1-NN without transformation

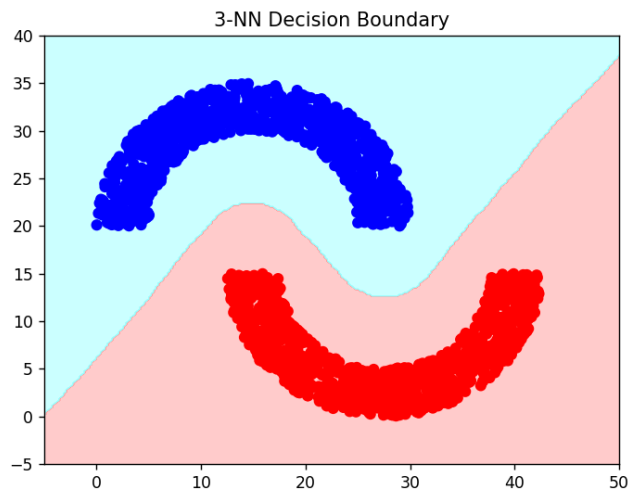


Figure 6: 3-NN without transformation

5 Problem 6.16

(a)

Two algorithms, the 10-partition branch and bound and brute-force search, were run on 10,000 randomly generated test points with uniform data distribution. The brute-force search took approximately 153.49 seconds, while the branch and bound method completed in just 15.66 seconds. This significant difference demonstrates that branch and bound is much faster due to its ability to prune large sections of the search space, effectively skipping unnecessary points and focusing on relevant sections.

(b)

For a Gaussian distribution of points, the experiment was repeated under similar conditions. The brute-force search time remained close to the previous result, taking around 158.67 seconds. However, the branch and bound algorithm improved slightly, taking only 18.06 seconds. This improvement in branch and bound's performance with Gaussian data likely results from the more structured clustering in Gaussian data, allowing the branch and bound algorithm to optimize its search further by focusing on relevant clusters.

(c)

The observations in parts (a) and (b) underscore the advantages of branch and bound over brute-force search. Brute-force requires scanning the entire dataset, which is time-intensive for large datasets like the 10,000 points here. In contrast, branch and bound leverages partitioning, which groups points based on their relative distances, allowing it to skip large sections of the data when searching for the nearest neighbor.

(d)

The number of test points generally does not change the optimal choice of algorithm. Branch and bound tends to outperform brute force regardless of dataset size, as it efficiently skips irrelevant points. Furthermore, when data is well-clustered, as in Gaussian distributions, the branch and bound algorithm performs even better due to the naturally grouped data points.